

Once
you stop
learning,
you start
dying.

—Albert Einstein

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breakdown, outperforming other semiconducting materials. Diamond materials have extended the application range of power devices.

The Baliga figure of merit (BFOM), frequently used to assess the effectiveness of power devices, is defined as follows:

$$\text{BFOM} = \text{BV}^2 / R_{\text{on}}$$

where BV represents the breakdown voltage and R_{on} refers to the specific on-resistance. The device's conduction power loss and off-state performance at low frequencies are represented by the BFOM. The higher the BFOM value, the greater the off-state reverse voltage the device can sustain, and the higher the conductivity per unit area when activated. The remarkable potential of diamond power semiconductors for high-voltage and high-power applications is emphasised by the high BFOM. Diamond power devices, with their outstanding BFOM, are poised to become essential components for next-generation electronics in both military and civilian applications, a fact recognised in numerous countries.

Diamonds are a valuable material for electronic applications due to their exceptional properties as semiconductors and electrical insulators. However, the adoption of diamond in IC packaging has been gradual, although it has begun to be integrated into high-power, LED, and RF devices, as high-quality, thick diamond plates have only become achievable within the last five to ten years.

Applications of diamond semiconductors

Diamond semiconductor technology is advancing CMOS integrated circuits and enabling computation at higher speeds and in more extreme temperatures. Energy-efficient electronics, spintronic devices, and sensors based on

micro-electromechanical systems (MEMS) that perform in challenging environments, such as space, could all benefit from diamond semiconductors. These semiconductors could also find use in electric vehicles (EVs), supercomputers, and consumer devices that require enhanced robustness and reduced weight.

Recent estimates indicate that power electronics currently handle 50% of the world's electricity, and as the demand for electricity and energy storage systems increases, this percentage is predicted to grow to 80%. While the current generation of semiconductor power devices meets present-day needs, achieving carbon neutrality and supporting the transition from fossil fuels will require new devices capable of handling the rising electrification of modern society.

The power electronics industry, which has long depended on silicon, is undergoing a transformation influenced by the growing popularity of electric vehicles, accelerating the shift to new semiconductor materials with enhanced capabilities. Diamond is emerging as the preferred semiconductor material due to innovations in the transistor industry, resulting in lighter, more energy-efficient components that tolerate higher temperatures. These advances are expected to be crucial for various applications, including electrical distribution networks, large-scale industries, and power electronics in electric vehicles.

India's transistor industry is set to establish an initial production line, supported by funding from government or private organisations and backed by dedicated investors, to further develop this technology, expedite the pre-industrialisation phase, and address the increasing demand for diamond semiconductors. **EFY**